

# Reporte de Análisis Físico

## Validación MC y Asimetría UMD/SD

### Configuración de la Simulación

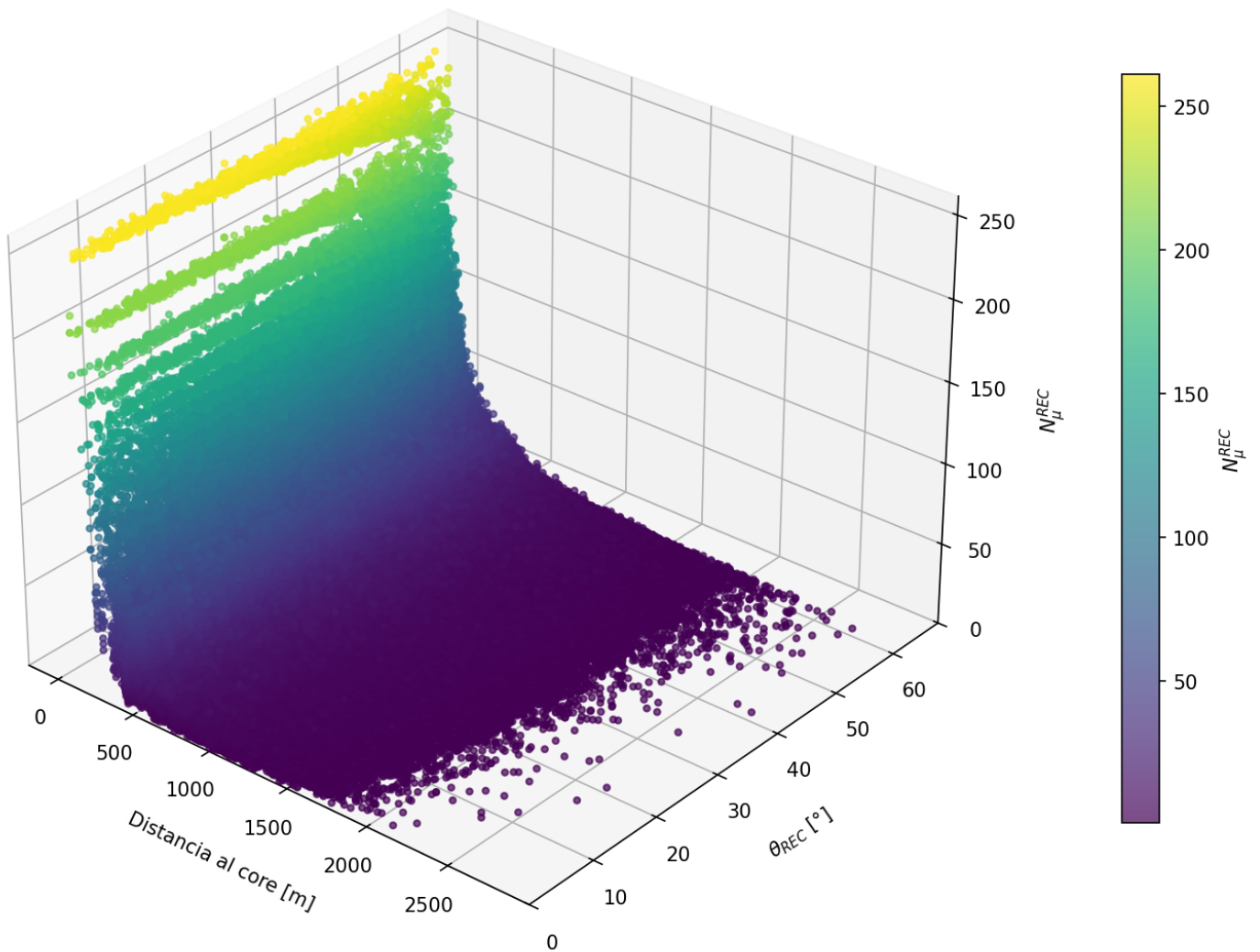
| Parámetro        | Valor               |
|------------------|---------------------|
| Modelo Hadrónico | EPOS-LHC            |
| Primario         | Helio               |
| Rango Energía    | 18.0 - 18.5 log(eV) |
| Carpeta          |                     |
| Eventos Totales  | 5,000               |
| Módulos Totales  | 2,227,440           |

## 1. Limpieza de Datos

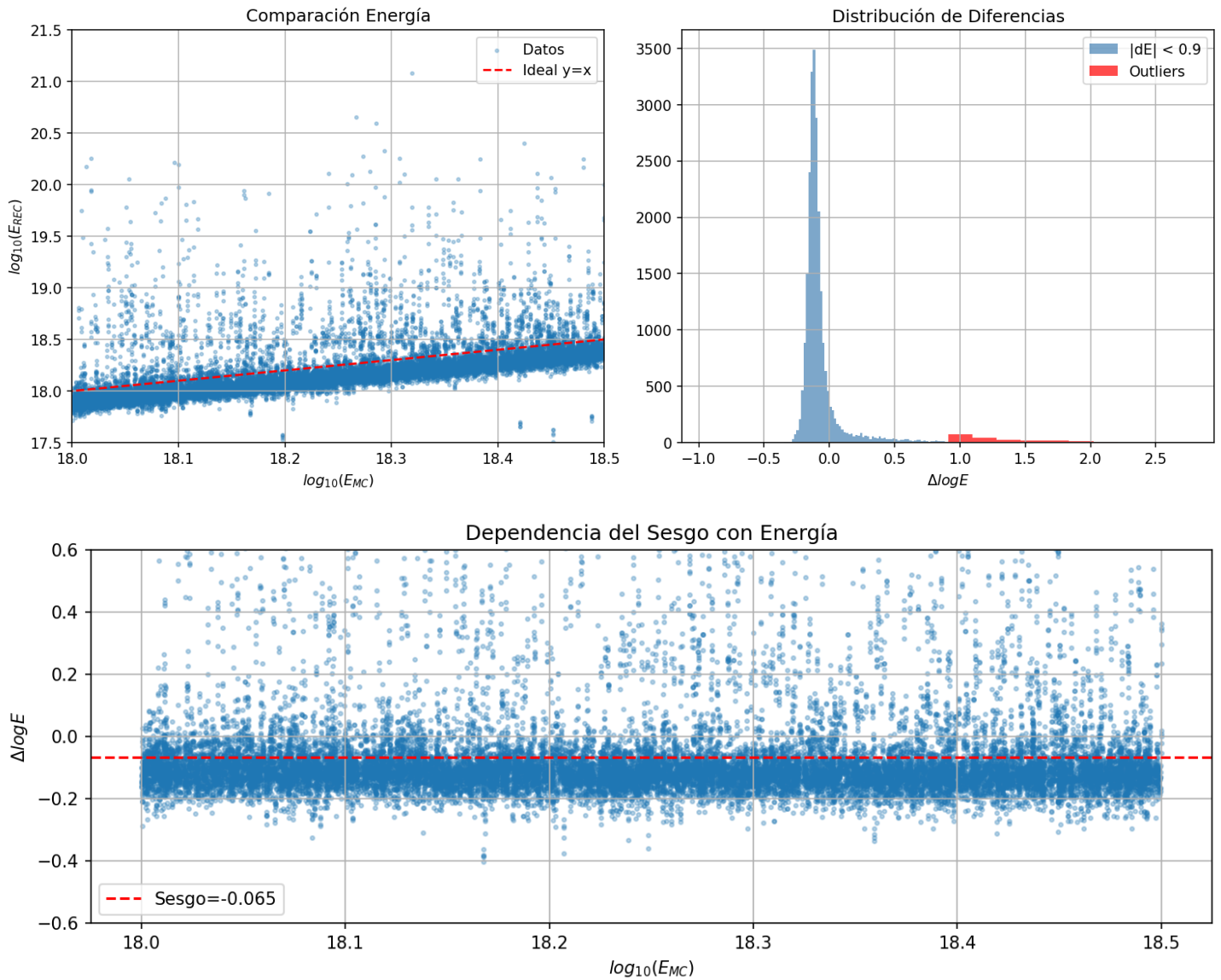
### REPORTE DE LIMPIEZA

| Métrica          | Valor                    |
|------------------|--------------------------|
| Filas Iniciales  | 2227440                  |
| Filas Eliminadas | 109164 (4.90%)           |
| Filas Finales    | 2118276                  |
| Causa Principal  | NaN en logE_REC (109164) |

## 2. Visualización General

$N_{\mu}^{REC}$  vs distancia y  $\theta$ 

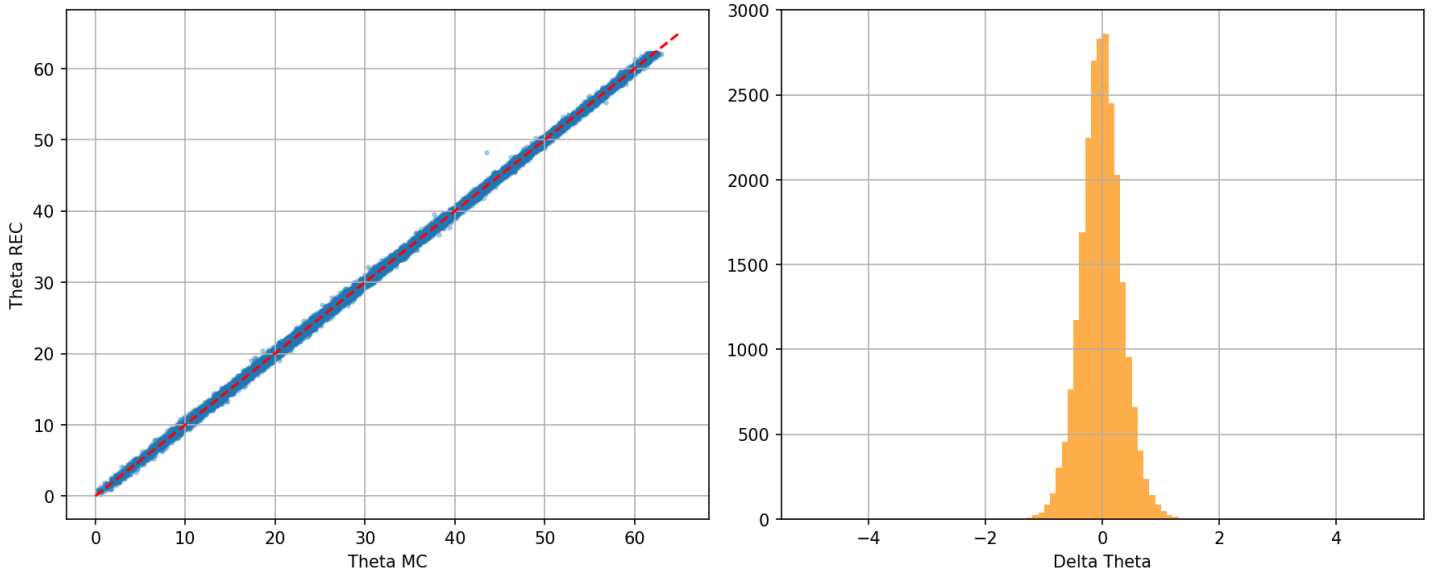
### 3. Resolución de Energía



Sesgo (Mean): -0.0654  
 Resolución (Std): 0.1941  
 Outliers (> 0.9): 200 (0.84%)

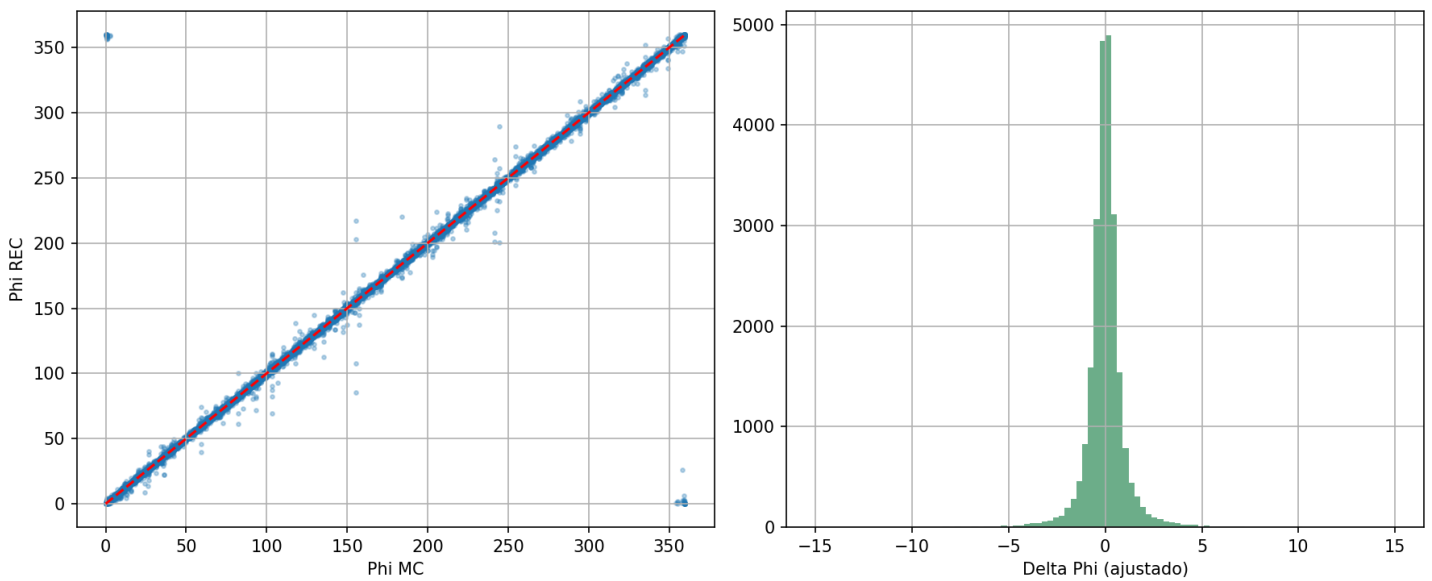
## 4. Resolución Angular

Theta (Cenital)



Sesgo:  $-0.0168$  deg  
Resolución:  $0.3504$  deg  
Outliers ( $> 10.0^\circ$ ): 0 (0.00%)

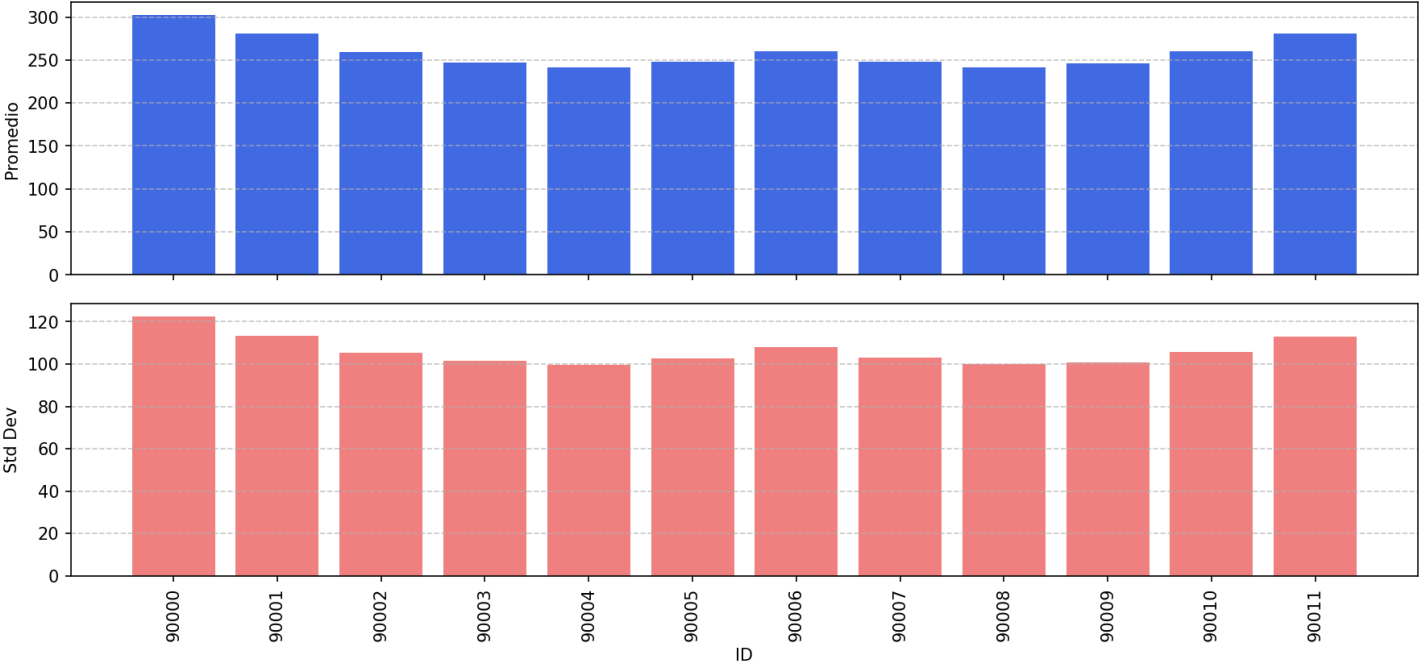
### Phi (Azimutal)



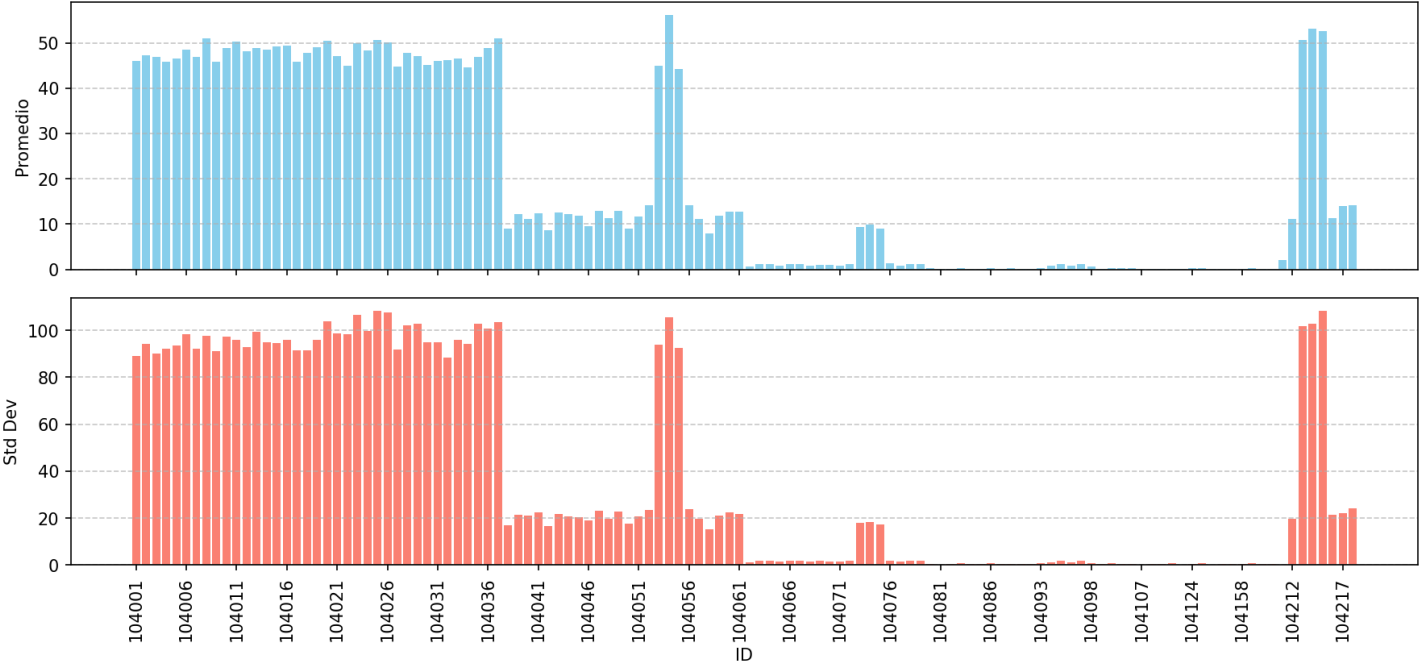
Sesgo (Median):  $0.0011$  deg  
Resolución (MAD):  $0.3865$  deg  
Outliers ( $> 15.0^\circ$ ): 32 (0.13%)

## 5. Uniformidad por Counter

Uniformidad: Anillo Denso (90k)

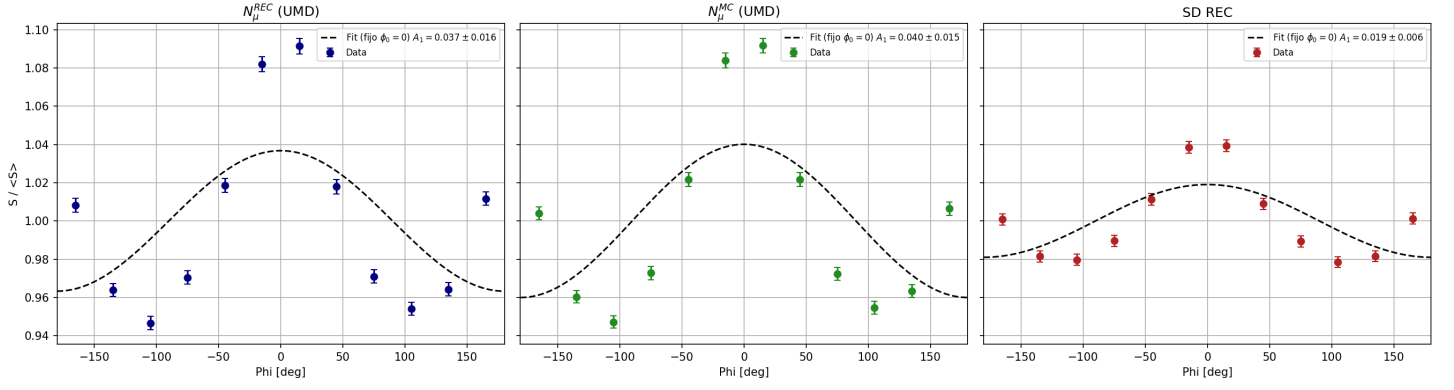


Uniformidad: Infill (104k)

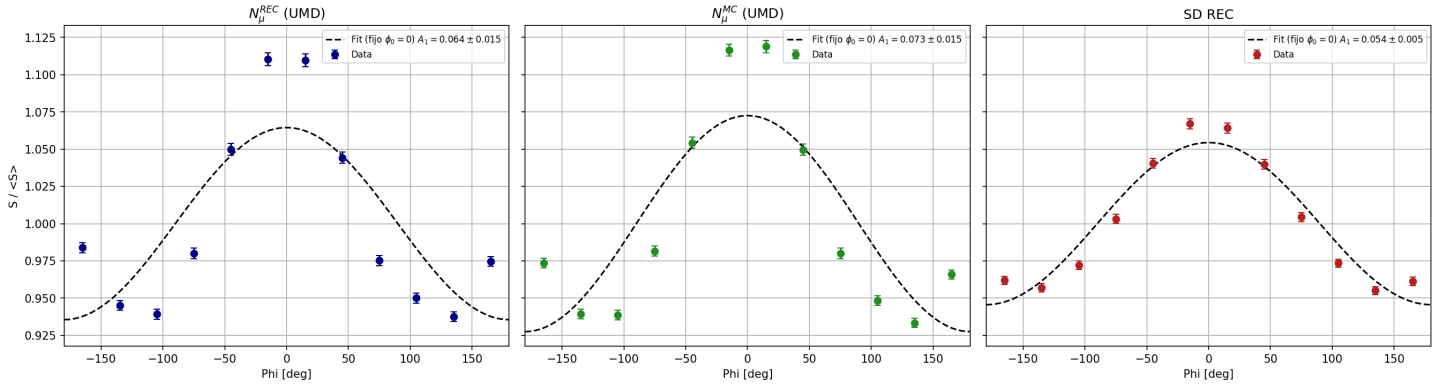


## 6. Asimetría Azimutal

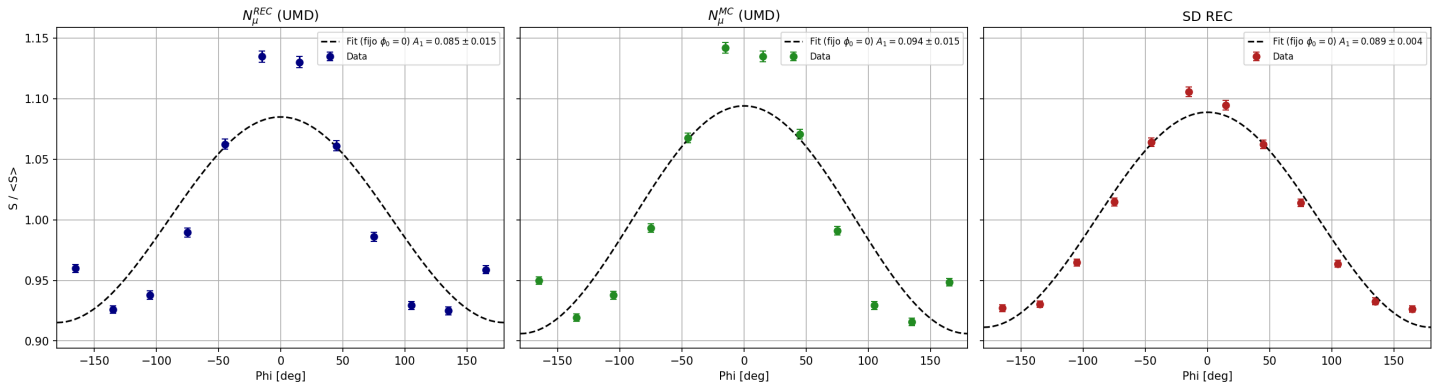
Bin 0: Theta 0.3 - 21.2 deg



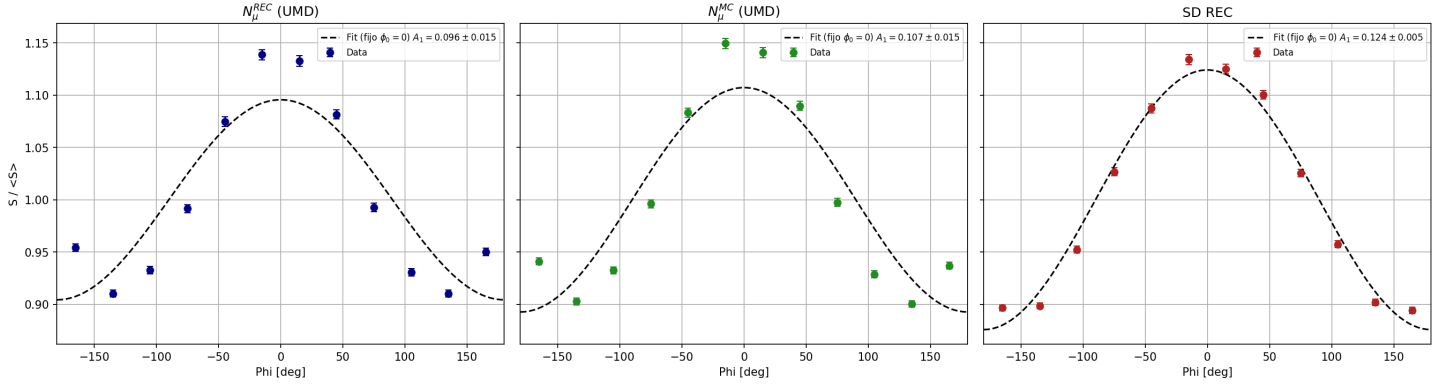
### Bin 1: Theta 21.2 - 30.7 deg



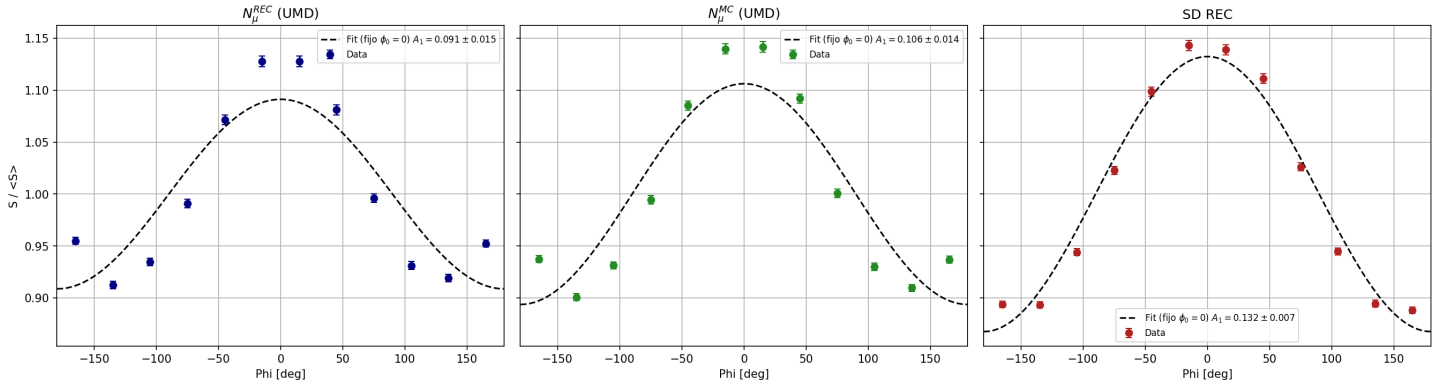
### Bin 2: Theta 30.7 - 38.7 deg



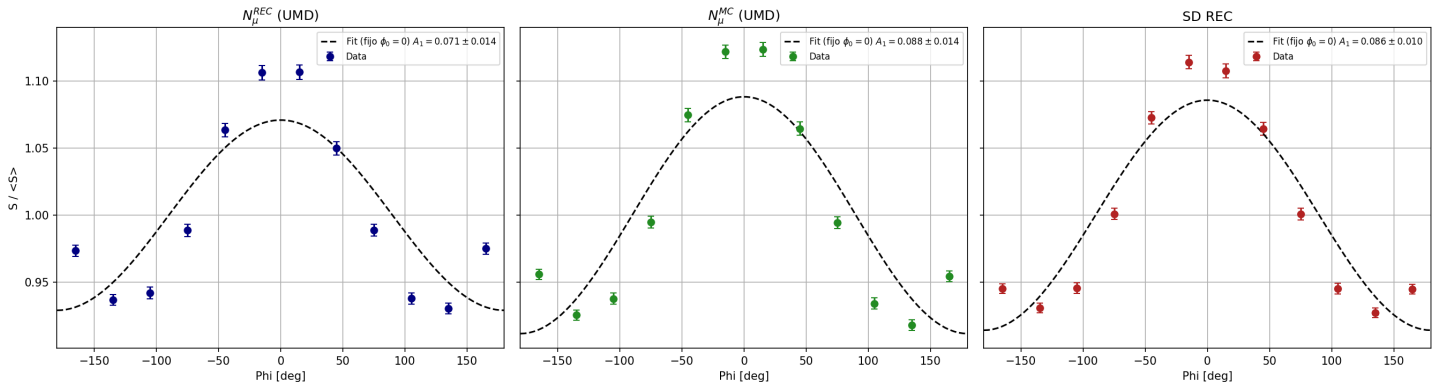
### Bin 3: Theta 38.7 - 46.2 deg



#### Bin 4: Theta 46.2 - 53.8 deg



#### Bin 5: Theta 53.8 - 62.2 deg



## 7. Resumen de Ajustes (Comparación)

Resultados Detallados: Amplitud A1 y Error

| theta_range | A1_UMD_REC | Err_UMD_REC | A1_UMD_MC | Err_UMD_MC | A1_SD_REC | Err_SD_REC |
|-------------|------------|-------------|-----------|------------|-----------|------------|
| 0.3-21.2    | 0.0367     | 0.0157      | 0.0400    | 0.0152     | 0.0190    | 0.0064     |
| 21.2-30.7   | 0.0645     | 0.0150      | 0.0725    | 0.0145     | 0.0544    | 0.0046     |
| 30.7-38.7   | 0.0849     | 0.0152      | 0.0940    | 0.0145     | 0.0888    | 0.0044     |
| 38.7-46.2   | 0.0957     | 0.0154      | 0.1073    | 0.0148     | 0.1242    | 0.0051     |
| 46.2-53.8   | 0.0913     | 0.0146      | 0.1063    | 0.0141     | 0.1324    | 0.0068     |
| 53.8-62.2   | 0.0709     | 0.0141      | 0.0883    | 0.0137     | 0.0858    | 0.0101     |

